

TARNAVENI, Romania

WMO: 151650

Lat: 46.3600N

Lon: 24.2261E

Elev: 1720

StdP: 13.81

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	14.9	4.2	7.3	12.2	9.4	9.5	18.2	30.9	33.9	26.4	31.5	3.9	340	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.4	85.8	67.3	83.2	66.5	80.6	65.8	69.8	81.3	68.4	79.6	67.2	77.8	5.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	101.8	74.7	64.7	97.5	73.4	63.4	93.2	72.0	34.9	81.0	33.7	79.5	32.6	78.0	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	22.1	18.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	28.1	34.5	41.3	50.7	59.8	66.5	68.1	69.7	60.7	51.3	41.9	31.8
	DBStd	16.08	8.39	8.43	8.38	9.47	6.49	6.18	5.43	6.09	8.24	7.82	8.75	7.32
	HDD50	2438	678	437	287	107	8	0	0	0	6	84	266	565
	HDD65	5788	1143	855	734	431	184	56	32	27	178	426	693	1029
	CDD50	2600	0	2	19	128	311	494	561	611	328	123	23	0
	CDD65	477	0	0	0	3	22	100	128	173	50	1	0	0
	CDH74	3023	0	0	0	16	115	582	747	1260	301	2	0	0
	CDH80	686	0	0	0	0	8	110	162	345	61	0	0	0
Wind	WSAvg	7.2	6.7	8.1	8.5	9.3	7.6	7.1	6.5	5.7	6.9	6.8	6.9	6.7
Precipitation	PrecAvg	22.4	0.9	0.7	1.1	1.9	2.7	3.7	3.2	1.9	2.2	1.8	1.3	1.2
	PrecMax	30.3	1.7	1.6	2.2	3.8	4.3	8.4	5.0	3.6	5.8	3.5	2.8	2.5
	PrecMin	15.5	0.2	0.1	0.3	0.1	1.0	1.0	0.9	0.2	0.2	0.1	0.0	0.1
	PrecStd	4.2	0.5	0.4	0.6	0.9	1.0	1.6	1.3	1.0	1.2	1.0	0.6	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	57.4	67.5	76.3	80.8	86.2	87.4	90.3	85.7	72.7	64.4	50.7
		MCWB	43.6	48.3	51.6	57.5	61.5	69.9	69.6	68.4	64.8	58.3	53.1	45.0
	2%	DB	44.8	52.9	61.5	72.7	77.3	83.2	84.2	86.6	80.7	69.6	60.4	47.5
		MCWB	41.3	45.7	48.6	55.8	60.9	67.6	67.8	66.9	63.7	57.0	51.4	43.2
	5%	DB	41.7	49.6	57.0	68.8	74.4	80.1	81.2	83.8	77.0	66.9	57.2	44.7
		MCWB	38.8	43.8	45.9	53.9	59.8	66.8	66.9	66.1	61.4	55.4	50.3	41.0
	10%	DB	38.8	46.1	52.8	64.7	71.0	77.2	78.2	80.9	73.4	63.7	54.0	41.6
		MCWB	36.5	41.7	43.6	52.1	58.3	65.2	65.7	65.8	60.5	54.3	48.4	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.0	50.1	52.6	59.2	64.7	72.8	71.1	70.8	67.3	60.1	54.0	46.5
		MCDB	47.2	55.8	66.2	72.3	76.7	83.5	83.2	84.3	81.6	68.8	61.7	48.9
	2%	WB	42.0	46.4	49.3	57.2	62.9	69.8	69.5	69.3	64.9	58.2	52.3	43.5
		MCDB	44.7	51.6	60.1	69.7	73.9	80.0	80.7	81.7	77.3	67.3	58.8	47.4
	5%	WB	39.4	44.1	46.9	55.3	61.1	67.9	68.2	67.9	63.0	56.5	50.9	41.1
		MCDB	42.0	48.9	55.8	66.6	71.4	78.1	78.5	80.4	74.3	65.0	56.7	43.9
	10%	WB	36.7	42.0	44.7	53.3	59.5	66.1	66.9	66.7	61.6	54.9	49.0	39.1
		MCDB	38.5	46.1	51.3	63.2	68.8	75.5	76.0	78.3	71.4	62.3	54.2	41.6
Mean Daily Temperature Range	5% DB	MDBR	7.8	10.6	13.2	15.9	15.3	16.0	15.9	17.4	15.5	13.5	9.7	6.9
		MCDBR	10.1	15.4	18.7	20.7	19.2	19.0	19.3	20.6	20.0	19.1	13.6	10.8
		MCWBR	7.4	9.4	10.3	9.6	8.3	8.4	7.7	7.2	7.4	8.9	7.3	8.0
	5% WB	MCDBR	9.8	14.5	17.4	19.0	17.6	18.6	17.6	19.2	18.9	17.1	11.7	10.4
MCWBR		7.5	9.4	9.8	9.2	8.4	8.8	7.9	7.4	7.7	8.2	6.8	7.8	
Clear-Sky Solar Irradiance	taub	0.305	0.331	0.370	0.420	0.404	0.400	0.419	0.425	0.386	0.353	0.331	0.307	
	taud	2.351	2.315	2.268	2.155	2.260	2.298	2.249	2.230	2.310	2.393	2.406	2.377	
	Ebn at Noon	249	264	269	265	274	275	267	260	261	253	237	234	
	Edh at Noon	23	30	36	45	42	40	42	41	35	28	23	21	
All-Sky Solar Radiation	RadAvg	408	673	1026	1379	1691	1843	1824	1676	1196	821	467	315	
	RadStd	53	78	126	140	156	157	155	152	132	100	59	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (76 neighbors)	+1.33	N/A	N/A	+1.78	N/A	N/A	-274	-396	+166	+34

Nomenclature: See separate page